4. Inner, darker & harder portion of secondary xylem that cannot conduct water in an older dicot stem is called
   a. Alburnum b. Bast
   c. Wood d. Duramen
5. What is the fate of primary xylem in a dicot stem showing extensive secondary growth?
   a. It is retained in the centre of the axis
   b. It gets crushed
   c. May or may not get crushed
   d. It gets surrounded by primary phloem

Assertion and Reason

6. These questions consist of two statements each printed as Assertion and Reason. While answering these questions you are required to choose any one of the following four responses.
   A. If both Assertion and Reason are true but the Reason is a correct explanation of the Assertion.
   B. If both Assertion and Reason are true but Reason is not a correct explanation of the Assertion
   C. If Assertion is true but the Reason is false.
   D. If both Assertion and Reason are false.
   1. **Assertion:** In woody stems the amount of heart wood continue to increase year after Year
      Reason: The activity of cambial ring continues uninterrupted
      a. A b. B c. C d. D
   2. **Assertion:** Secondary growth in dicot roots occurs with the help of vascular cambium and phellogen.
      Reason: Vascular cambium is completely primary in origin.
      a. A b. B c. C d. D

Answer the Following

7. Explain the activity of successive cambium with example.
8. What is wood botanically?
9. In a forest, if the bark of a tree is damaged by the horn of a deer, How will the plant overcome the damage?
10. Differentiate the wood formed in *Pinus* from that of *Morus*.
11. In which season the vessels of angiosperms are larger in size, why?
12. Central part of the wood is always dark. Why?
13. Continuous state of dividing tissue is called meristem. In connection to this, what is the role of lateral meristem?
14. A timber merchant bought 2 logs of wood from a forest & named them A & B, The log A was 50 year old & B was 20 years old. Which log of wood will last longer for the merchant? Why?
15. A cross section of tree trunk contains 60 lighter and 60 darker rings. Determine the age of the tree and justify.
16. A transverse section of the trunk of a tree shows concentric rings which are known as growth rings. How are these rings formed? What are the significance of these rings?
17. There are many tissues produced outside the vascular cambium in the stem. Explain them.
18. When you go to a timber mart to collect timber for your construction of a new house, how will you select good timber?
19. Explain artificial seasoning of woods.

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